Activity 1, Advanced - Intro to Spatial Mapping

Activity 1, Advanced - Intro to Spatial Mapping

Download this activity

In this adventure, students will begin mapping the SVG Canvas units to physical and tangible drawing surfaces utilizing tactile drawing paper. Students will draw basic shapes on tactile dot or graph paper and understand how to mark and understand shape positions around the canvas and measure the dimensions of their shapes.

Setup - Activity 1

This activity can be broken down into several elements for lesson planning.

Objectives

- Learn how a drawing on paper can be mapped into digital units by presenting a tactile coordinate system.
- Teaching that the origin is the top-left corner of any drawing in either portrait or landscape orientation and is represented as (0,0) or x=- and y=0.
- Teaching that we can think of the relative units as 100 units per inch, and that each dot on the provided dot paper or each line in the graph paper represents 50 unit increments across the paper.
- Have fun creating designs with basic shapes and understanding how to feel and note their dimensions/sizes using the tactile markings on the paper.

Materials Needed

- APH Draftsman Tactile Drawing Board or APH TactileDoodle
- Tactile Dot or Graph paper (Provided in TADA materials)
- Stylus or Pen
- Tactile Rulers
  - Cardstock or Cardboard/ for a straight edge that can be folded

Alternate Materials

- Sensational Blackboard

Ideas for Carrying Out This Activity

- Use basic shapes at first, offer up suggestions like making a circle 150 units in diameter, squares and rectangles of various sizes,and try connecting various dots together with lines to make triangles and polygons.
- If a student wants to progress further than basic shapes, encourage them to build a scene using various shapes and straight lines.

Adventure Map: Activity 1

Teaching tip: Provide sufficient time for the student to explore, develop skills, and have fun at each step! Encouraging creativity and personal preferences for drawing as much as possible. Some students may be able to accomplish each step in one session; most students will need several sessions to complete the adventure.

1. Introduction

By this point through TADA, students should be comfortable drawing basic shapes via tactile drawing surfaces. Ask your students how they think computers know how large or small to render graphics and where to place them on the screen or paper.

The answer is numbers, specifically the numbers the students give the computer. When drawing shapes, ideas like size, positioning, line thickness, and more are understood by the computer as "attributes" and the numbers that change these attributes are called "values." We have to know the values of the various attributes of the shapes we draw in order to accurately input them on the computer.

2. Create Your Graphic

Provide your student with drawing materials. Ask your student to draw a series of basic shapes with different attribute values on the tactile paper. The following are some example instructions:

- "Draw a circle that is 150 units in diameter in the center of the paper"
- "draw a rectangle 200 units wide and 50 units tall, starting the top-left corner of the rectangle 100 units over on X and 300 units down on Y,"
- "Draw a right triangle with the top point starting at 50 units over on X and 50 units down on Y, the second point 50 units over on X and 200 units down on Y, and the third point 250 units over on X and 200 units down on Y, and connect all the dots together."

Once students get an understanding of how to think about the attributes and the values as they appear on the tactile paper, have them explore drawing their own scene or set of shapes, sticking to basic shapes and straight lines.

3. Talking through Design Process

- Have students speak out their shape attributes and values.
- Have students toss out shapes and values to the other students to try drawing on their own workspaces.
- Tie this back to the previous lessons about communicating drawing instructions.

4. Conclusion

Once students are comfortable with the canvas space, get them excited for the next step of translating their designs into actual code on the computer that will render it digitally.

- Does feeling their drawings and knowing the dimensions make them feel more assured about the numbers they'll be using to build their shapes?
- Does the canvas system make sense?
- Reinforce that the top-left corner of the drawing is (0, 0) and is known as the origin, and that for a standard 8.5"x11" sheet of Letter paper in landscape orientation, this means that the top-right corner can be represented as (1100, 0), the bottom-right corner as (1100, 850), and the direct center of the page as (550, 425).

Download the whole thing · **Next:** Activity 2: Digitizing Drawings with SVG Code · **Back to:** Adventure Map